

Three questions on branching Markov chains

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Goal: Understanding how branching random walks in random environment and catalytic branching random walks spread out.

Outline

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- 1 The model: branching random walk/branching Markov chain
- 2 Recurrence/Transience
- 3 The maximal distance to the origin
- 4 Localisation for branching Markov chains with a catalyst

Consider a branching Markov chain in discrete time on a (countable) state space S , starting with one particle at the origin (some fixed state $0 \in S$). It is given by two ingredients:

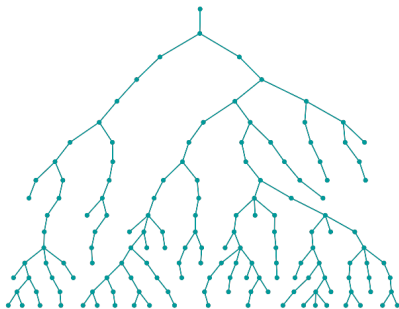
- An offspring distribution μ on $\mathbb{N}_0$ with weights $(\mu(k))_{k \in \mathbb{N}_0}$.
In this talk, for simplicity, assume $\mu(0) = 0$.
- An irreducible Markov chain on S , given by transition matrix P i.e. transition probabilities $p(x, y)$, $x, y \in S$.

At time 0 the process starts with one particle at the origin. At every time $n \in \mathbb{N}$ each particle repeats the following two steps, independently of everything else. First, it produces offspring according to the offspring distribution μ , and then it dies. Afterwards, the offspring particles move independently according to the Markov chain.

$$m := \sum_{k=0}^n kp(k)$$

is the reproduction mean of the offspring distribution. Will assume $m > 1$. Let Z_n be the number of particles in generation n . Then, (Z_n) is a supercritical branching process.

The branching Markov chain can be interpreted as a tree-indexed Markov chain:



Let $\mathcal{T} = (V, E)$ be the genealogical tree corresponding to (Z_n) . Let D_n be the set of vertices in the n -th generation of the tree. For $v \in D_n$ the position S_v of the particle v at time n is given by the state of the Markov chain, started from $0 \in S$ and run for n steps. Of course, for $v, w \in D_n$, S_v and S_w are in general not independent.

Example (BRW with drift)

Let $p \in [\frac{1}{2}, 1)$ and consider the Markov chain on $S = \mathbb{Z}$ given by

$$p(x, y) = \begin{cases} p & y = x + 1, \\ 1 - p & y = x - 1. \end{cases}$$

This is a branching random walk.

Example (Variation in random environment)

Let $(\omega_x)_{x \in \mathbb{Z}}$ be an i.i.d. sequence with values in $[\frac{1}{2}, 1)$. Given $(\omega_x)_{x \in \mathbb{Z}}$, consider the Markov chain on $S = \mathbb{Z}$ given by

$$p(x, y) = \begin{cases} \omega_x & y = x + 1, \\ 1 - \omega_x & y = x - 1. \end{cases}$$

Recurrence/Transience

Question (Recurrence/Transience)

Is 0 occupied infinitely often by a particle?

Theorem (Russell Lyons/Robin Pemantle/Yuval Peres, NG/ Sebastian Müller for the critical case $m = 1/\rho$)

(i)

$$\mathbb{P}[0 \text{ is occupied infinitely often}] = \begin{cases} 0 & \text{if } m \leq 1/\rho, \\ > 0 & \text{if } m > 1/\rho, \end{cases}$$

where

$$\rho := \limsup_{n \rightarrow \infty} P^n(0, 0)^{1/n} \in (0, 1] \quad (= e^{-I(0)}).$$

is the spectral radius of the Markov chain.

(ii) If the Markov chain is transitive (think of S as a directed graph with edges from x to y if $p(x, y) > 0$), then

$$\mathbb{P}[0 \text{ is occupied infinitely often}] = 1 \text{ if } m > 1/\rho.$$

Going back to the examples: for the BRW with drift, $\rho = \sqrt{4p(1-p)}$, hence

$$\mathbb{P}[0 \text{ is occupied infinitely often}] = \begin{cases} 0 & \text{if } m \leq \frac{1}{\sqrt{4p(1-p)}} \\ 1 & \text{if } m > \frac{1}{\sqrt{4p(1-p)}}. \end{cases}$$

For the variation in random environment, $\rho = \sqrt{4\bar{p}(1-\bar{p})}$ where $\bar{p} = \min\{u : u \in \text{supp}(\omega_0)\}$.

Proof

Key observation: the particles that are at 0 form a branching process!
If $m < 1/\rho$, this branching process is subcritical:

$$\mathbb{E} \left[\sum_{v \in D_k} \mathbf{1}_{\{S_v=0\}} \right] = m^k \cdot P^k(0,0)$$

hence

$$\mathbb{E}[|total\ progeny|] = \sum_{k=0} m^k \cdot P^k(0,0) < \infty \quad \text{since } m < 1/\rho.$$

Proof

(continued)

A similar argument shows that if $m > 1/\rho$, the branching process is supercritical: for k large enough,

$$\mathbb{E} \left[\sum_{v \in D_k} \mathbf{1}_{\{S_v=0\}} \right] = m^k \cdot P^k(0,0) > 1.$$

With a bit more work, can show that in the case $m = 1/\rho$, the branching process is critical.

Conjecture

(ii) in the theorem should still be true if the Markov chain is “statistically transitive”. For such Markov chains, one should have

$$\mathbb{P}[0 \text{ is occupied infinitely often}] = \begin{cases} 0 & \text{if } m \leq 1/\rho, \\ 1 & \text{if } m > 1/\rho, \end{cases}$$

Confirmed for random walks in random environments on $\mathbb{Z}^d$ (Francis Comets and Serguei Popov, and S. Müller) and for random walks on Bienaymé-Galton-Watson trees (J. Berestycki, NG, D. Geldbach, Q. Shi).

The maximal distance to the origin

For $n \in \mathbb{N}$ consider the maximal distance of the particles to the origin at time n

$$R_n = \max_{v \in D_n} |S_v|,$$

where $|S_v|$ is the distance of the particle v to the origin (in the directed graph given by the Markov chain, with vertex set S). How does R_n grow?

Remark

If $S = \mathbb{Z}$, can look at the rightmost/leftmost particles at time n .

This question is very well understood for branching random walk on $\mathbb{Z}$ or on $\mathbb{R}$, less well understood for branching random walk on $\mathbb{Z}^d$ or on $\mathbb{R}^d$ or in random environment.

Assume that $|S_n|/n$ satisfies a large deviation principle with rate function I . Let $x^* := \sup\{u : I(u) \leq \log m\}$.

Theorem (John D. Biggins, John M. Hammersley, John F. C. Kingman)

(i)

$$\limsup_{n \rightarrow \infty} \frac{R_n}{n} \leq x^*.$$

(ii) *If the Markov chain is transitive, (or for branching random walk on $\mathbb{R}^d$),*

$$\liminf_{n \rightarrow \infty} \frac{R_n}{n} \geq x^*.$$

Indeed, the limit is the same as for the maximum of a (random number of) i.i.d. realizations of the Markov chain.

Conjecture

(ii) *in the theorem should still be true if the Markov chain is “statistically transitive”. For such Markov chains, one should have*

$$\lim_{n \rightarrow \infty} \frac{R_n}{n} = x^* . \quad (1)$$

Confirmed for random walks in random environments on $\mathbb{Z}$ (A. Callegaro, NG, C. Scali) and for (biased or simple) random walks on Bienaymé-Galton-Watson trees (J. Berestycki, NG, D. Geldbach, Q. Shi).

Question

If (1) holds, what is the order of $R_n - nx^$?*

Example (Random walk in random environment)

Let $(\omega_x)_{x \in \mathbb{Z}}$ be an i.i.d. sequence with values in $[\frac{1}{2}, 1)$ and $\text{Var}(\omega_0) > 0$. Given $(\omega_x)_{x \in \mathbb{Z}}$, consider the Markov chain on $S = \mathbb{Z}$ given by

$$p(x, y) = \begin{cases} \omega_x & y = x + 1, \\ 1 - \omega_x & y = x - 1. \end{cases}$$

Assume $x^* < 1$. Then, (A. Callegaro, NG, C. Scali), there is a (deterministic) $\sigma \in (0, \infty)$ such that

$$\mathcal{L} \left(\frac{R_n - nx^*}{\sigma \sqrt{n}} \right) \xrightarrow{w} N(0, 1).$$

More precisely, there is a sequence of random variables V_n^ω satisfying a CLT such that

$$|R_n - V_n^\omega| \leq C \log n \text{ almost surely for all } n \text{ large enough.}$$

Example (continuation)

This is at first sight surprising since in the deterministic case ($\text{Var}(\omega_0) = 0$)

$$\mathcal{L} \left(\frac{R_n - nx^*}{c \log n} \right) \xrightarrow{w} G.$$

where G is a Gumbel distribution.

Similar results were proved for branching random walks in *random branching* environments by Jiří Černý and Alexander Drewitz, and Xavier Kriechbaum.

Localisation for branching Markov chains with a catalyst

Consider a branching random walk in continuous time on $\mathbb{Z}^d$ with a catalyst at 0: particles at 0 produce offspring (with law μ) at rate $\lambda_0 \geq \lambda \geq 0$, particles not at 0 produce offspring (with law μ) at rate λ . Let $N(t)$ be the number of particles at time t . Let

$$\rho_n := \frac{1}{N(t)} \sum_{i=1}^{N(t)} \delta_{S_{v_i}} \in M_1(\mathbb{Z}^d)$$

be the empirical law of the particles alive at time t . We say that the BRW is **localised** if, almost surely, (ρ_n) converges weakly to a probability measure ν on $\mathbb{Z}^d$ and the BRW is **delocalised** if, almost surely, $\rho_n(A) \rightarrow 0$ for all finite $A \subset \mathbb{Z}^d$.

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The same definition applies to branching Markov chains (BMC).

Cécile Mailler and Bruno Schapira conjectured (and proved partially) (“A localisation phase transition for the catalytic branching random walk”, AAP 2026.)

Conjecture

The branching random walk described above is localised if and only if

$$\theta := (\lambda_0 - \lambda)(m - 1) > \gamma_d$$

(and delocalised otherwise).

Here γ_d is the escape probability of simple random walk on $\mathbb{Z}^d$.

Recall: m is the expectation of the offspring law.

The case $\lambda = 0$ was known before. See Sergio Albeverio, Leonid V. Bogachev, and Elena B. Yarovaya, 1998.

In joint work (in progress) with X. Chen, H. Hou and Q. Shi, we prove the conjecture, in a more general form for branching Markov chains (BMC) with a catalyst. We also discuss recurrence/transience for BMC with a catalyst.

Remark

Note that the BMC can not be localised if it is not recurrent. On the other hand, recurrence of the BMC does not imply localisation (consider simple random walk on $\mathbb{Z}^3$ and $\lambda_0 = \lambda$).

Questions are welcome and...

Thanks for your attention!